

NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

Higher 2

BIOLOGY

Paper 2 STRUCTURED QUESTIONS

9744/02

20th AUGUST 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [10]	
2 [11]	
3 [11]	
4 [10]	
5 [10]	
6 [8]	
7 [10]	
8 [10]	
9 [10]	
TOTAL P2 [30%]	100

This document consists of **22** printed pages and **0** blank pages.

[Turn over

Answer **all** questions.

- 1** Fig. 1.1 is an electron micrograph of structure A present in a mature organelle of a plant cell.

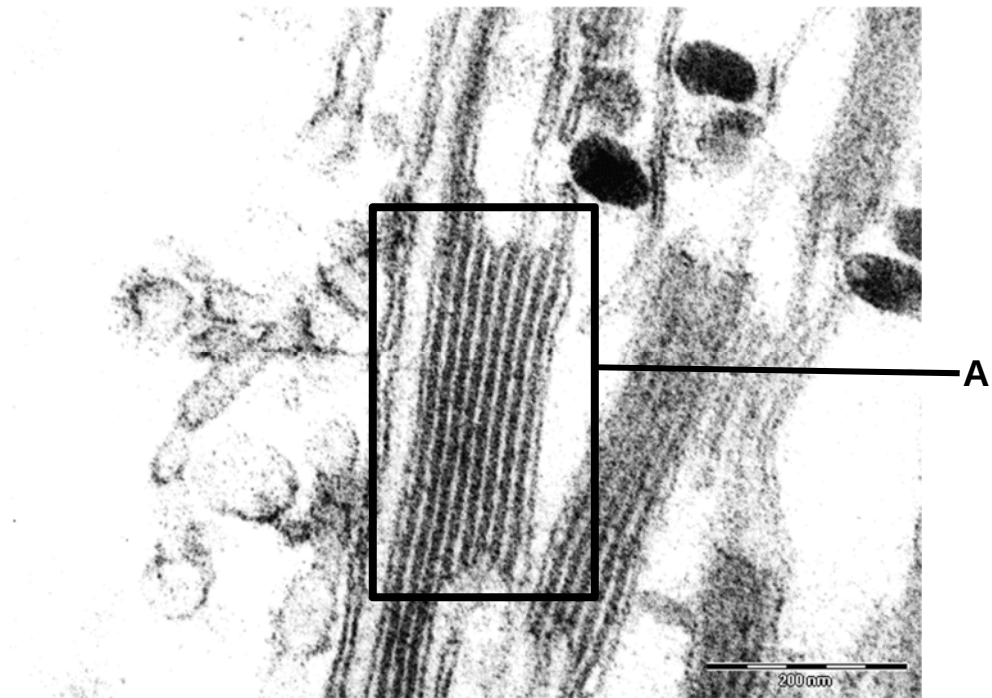


Fig. 1.1

- (a)(i)** Identify the organelle in which structure A resides.

.....[1]

- (a)(ii)** Explain the significance of the flattened stacks arranged in structure A..

.....

[2]

(b) Fig 1.2 shows a cell fractionation experiment performed by a student on a sample of plant cells.

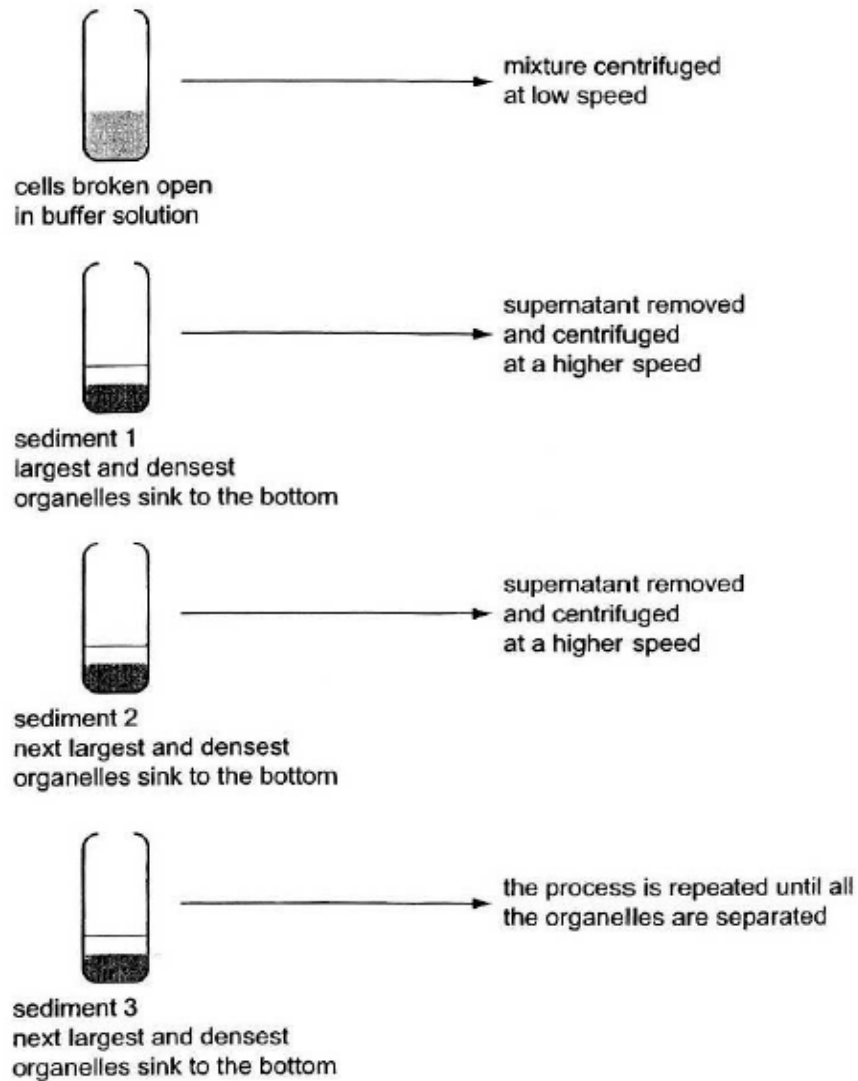


Fig. 1.2

However, she was not sure which sediment samples contained the organelles where structure A are located. Hence, she repeated the cell fractionation experiment again for all 3 sediment samples she collected, this time with a higher centrifugal speed.

Explain why there is a need for additional cell fractionation at a higher centrifugal speed.

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.....[2]

Fig. 1.3 shows a structure of a cell surface membrane.

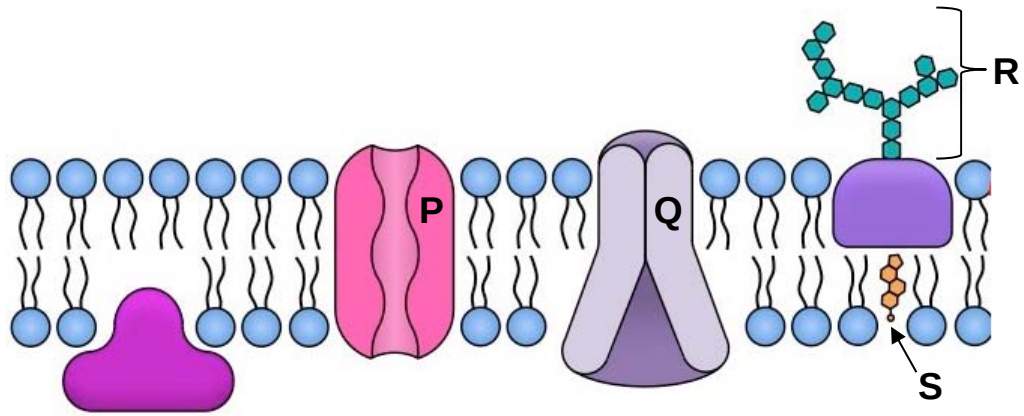


Fig. 1.3

With reference to Fig. 1.3,

(c)(i) Explain how the structure of Q allows it to be embedded in the cell surface membrane.

.....
[1]

(c)(ii) Explain how P functions differently compared to Q.

.....
[1]

(c)(iii) Explain why a change in R may cause an immunological response.

.....
[2]

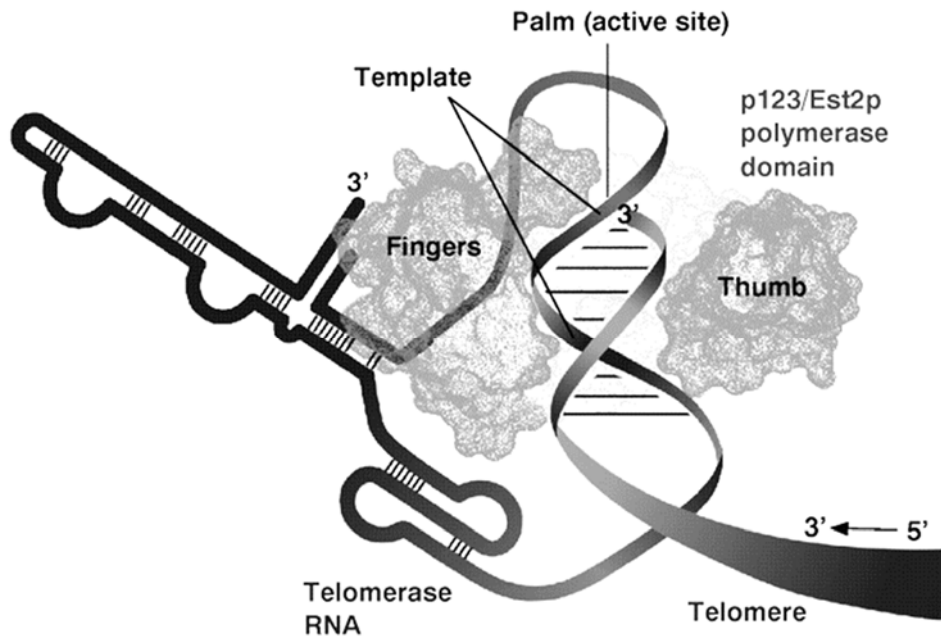
(c)(iv) Explain why structure S is amphipathic.

.....
[1]

[Total 10]

- 2 Telomerase is an important enzyme that lengthens telomeres and is extensively studied for its role in cancer. Molecularly, telomerase is a unique reverse transcriptase enzyme that carries its own RNA molecule.

Fig. 2.1 shows a commonly used “hand” model of the telomerase with its RNA template and a section of telomeric DNA bound.



Source: <http://science.sciencemag.org/content/276/5312/561.full>

Fig. 2.1

- (a) Telomerase is hypothesised to follow the induce-fit model in binding to DNA to carry out its function of repeatedly lengthening telomeric DNA. With reference to Fig 2.1, explain what is meant by the induce-fit model.

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.....

.....

.....[2]

The Telomere Repeat Amplification Protocol (TRAP) is a Polymerase Chain Reaction (PCR) based assay that is used to determine the activity of telomerases in cell lines quantitatively by measuring the amount of telomerase products in a given time. A graph of one such assay using HeLa cells (a type of cervical cancer cell line) as positive control is shown in Fig. 2.2.

Using the same technique, two other separate negative controls were set up where the results are shown in Fig. 2.3. One involved adding RNase into the mixture while the other involved heating up the HeLa cells.

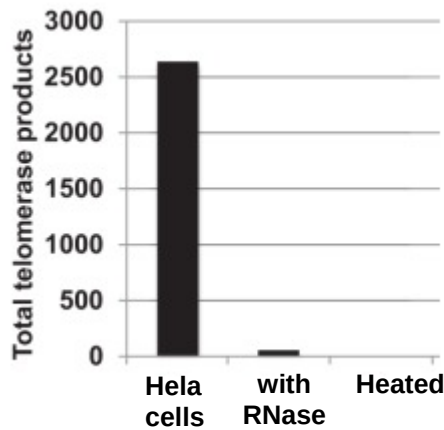


Fig. 2.3

- (c) When the cells were heated, there was no results. With reference to its structure in **Fig. 2.1**, explain why there was also very little to no result for when RNase was added to the cells culture.

.....[1]

- (d) Some of the more common treatments for cancer include different drugs that inhibit telomerases.

- (i) Explain why drugs such as telomerase inhibitors on their own are often ineffective in curbing cancer.

.....[2]

- (ii) Cancer patients sometimes develop resistance to their treatment, especially when there are tumours in different areas. Suggest why cancer cells of different tumours may exhibit different levels of resistance to the same drug.

.....[2]

[Total: 11]

- 3 Norepinephrine, also commonly known as noradrenaline, is a hormone released during stressful or dangerous situations eliciting fight-or-flight responses. The hormone binds to protein receptors on the cell membrane to exert its effects. Fig. 3.1 shows the molecular structure of norepinephrine.

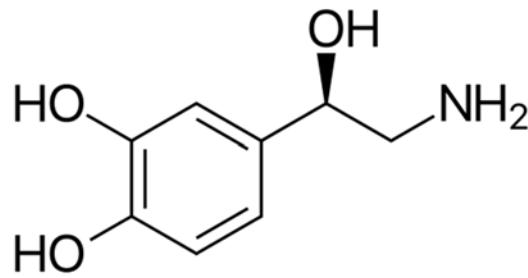


Fig. 3.1

- (a) With reference to Fig 3.1, explain why norepinephrine requires a protein receptor to exert its effects.

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.....[2]

- (b) Figure 3.2 shows an example of the cell signaling pathway taken by norepinephrine in eliciting cellular responses.

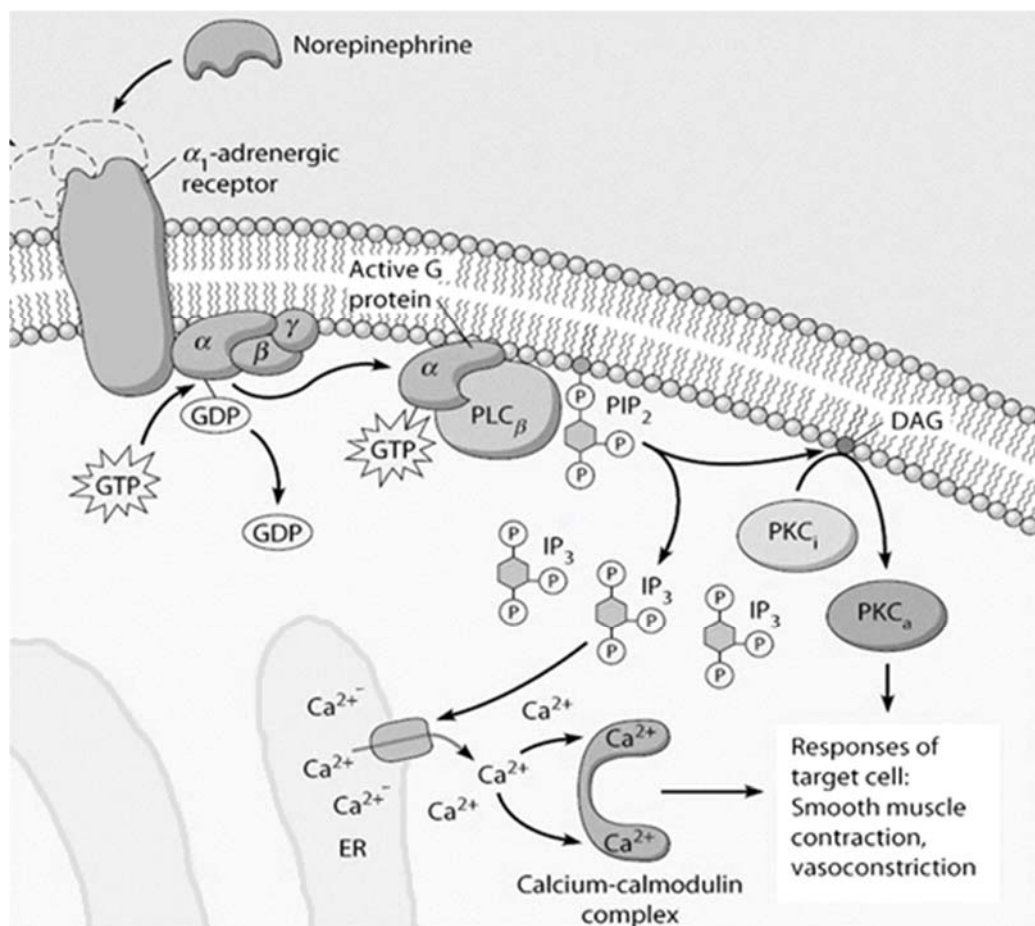


Fig. 3.2

- (i) Identify the type of receptor that norepinephrine binds to.

.....[1]

- (ii) With reference to Fig. 3.2, describe how an increased level of norepinephrine lead to the opening of IP_3 -gated Ca^{2+} channels.

.....

[4]

- (iii) With reference to **Fig. 3.2**, explain the significance of IP_3 and Ca^{2+} in the cell signalling pathway.

.....

[2]

Norepinephrine is produced during highly stressful fight-or-flight responses when an animal encounters a threat. It causes a number of effects such as increasing blood flow to the muscles and increasing alertness of the animal.

- (c) With reference to regulating stress levels in the body, explain why the production of norepinephrine by the body is also considered a form of homeostasis.

.....

[2]

[Total: 11]

- 4 Fig. 4.1 shows the potential application of Induced pluripotent stem cells (IPSCs). Stem cells, such as those from embryos, have been increasingly used as a solution to many diseases (e.g. diabetes) though with some resistance on ethical grounds.

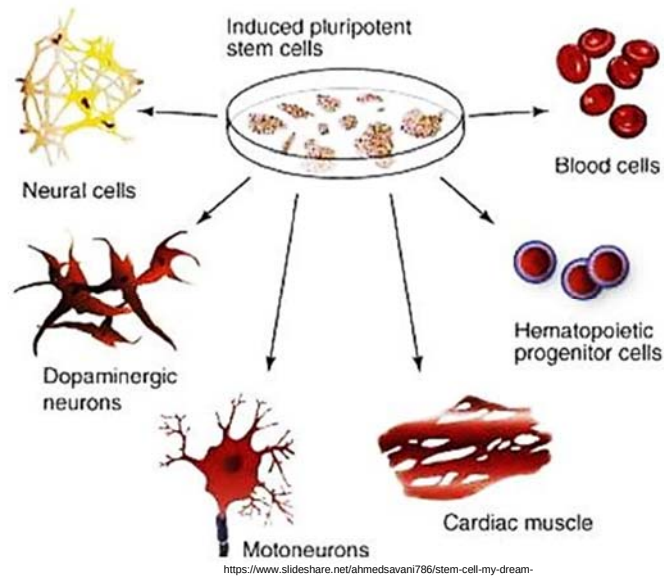


Fig. 4.1

- (a)(i) State and explain what ethical issues or considerations IPSCs overcome.

.....

.....

.....

.....[2]

- (a)(ii) With reference to Fig. 4.1, explain how the ability for IPSCs to proliferate is essential in the application of IPSC.

.....

.....[1]

- (b) With reference to Fig. 4.1, explain how the use of IPSCs overcome immunological challenges normally associated with previous stem cell therapies.

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.....[2]

- (c) With reference to Fig. 4.1, IPSC divide mitotically, explain which steps within the mitotic process ensure that IPSCs maintain their pluripotent capabilities.

.....

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.....

.....[2]

- (d)(i) In the conversion of differentiated cells to IPSC, it was discovered that several changes were needed at the chromatin level as part of the reprogramming into IPSCs.

Identify one such change and explain why it is needed.

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.....

.....

.....[2]

- (d)(ii) In relation to (d)(i), state one other modifications would be needed at the linear dna level.

.....

.....[1]

[Total: 10]

- 5 Fig. 5.1 shows the activation of transcription initiation in eukaryotes by recruitment of the eukaryotic RNA polymerase II holoenzyme complex

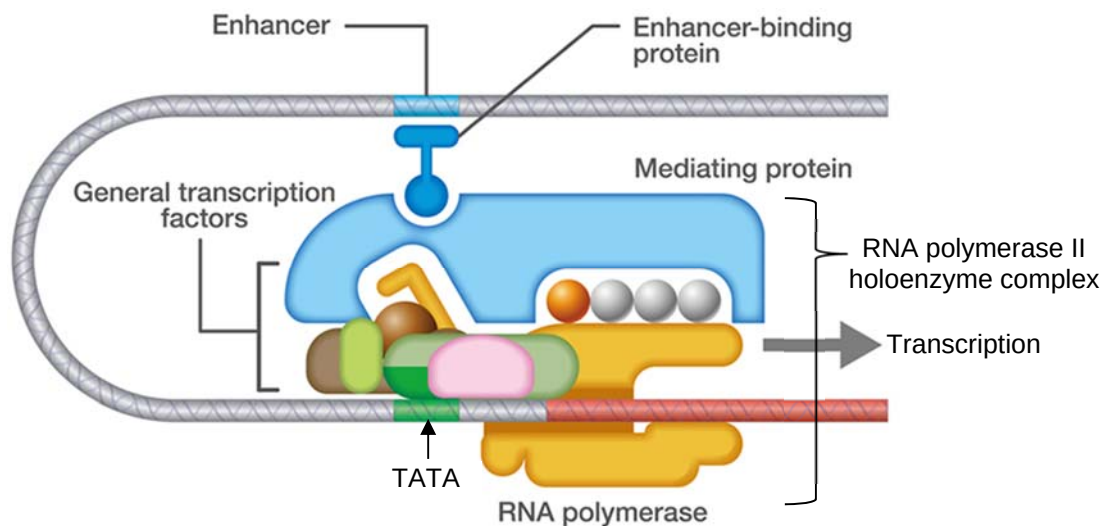


Fig. 5.1

- (a) With reference to Fig.5.1, identify a way in which transcription in eukaryotes differs from that in prokaryotes.

.....

.....[1]

- (b) Eukaryotic DNA contains both exons and introns. Fig. 5.2 shows the structure of a length of pre-mRNA formed from one gene.

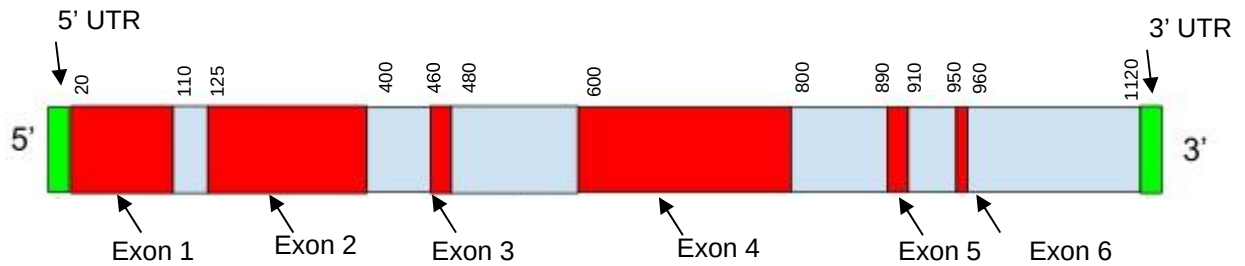


Fig. 5.2

- (i) Calculate the longest mRNA molecule that can be formed from the pre-mRNA shown in Fig. 5.2

.....
[1]

- (ii) Outline how this gene, shown in Fig. 5.2, could code for a number of different proteins.

.....
[1]

- (c) Eukaryotes and prokaryotes contain ribosomes, an integral cell structure responsible in protein synthesis. These ribosomes are made up of ribosomal RNA (rRNA) which is single stranded and unbranched polynucleotide chain. However, 70% of it is in the form of double stranded and helical. These double stranded regions form "hairpin loops", shown in Fig. 5.3.

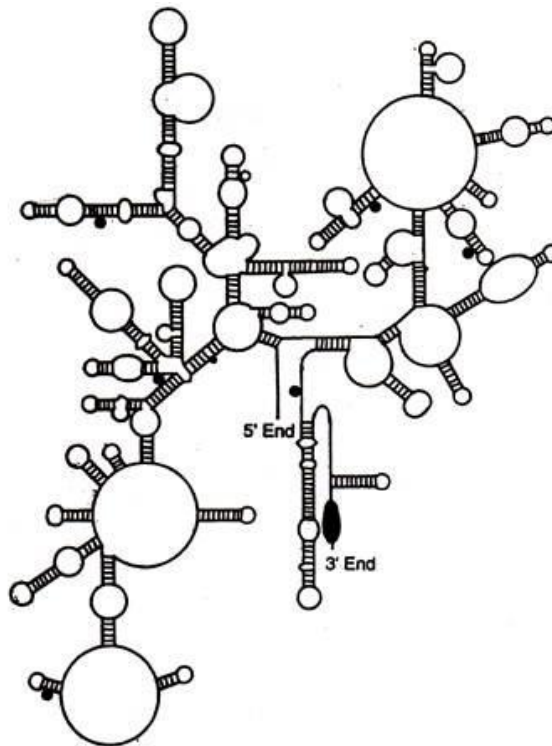


Fig. 5.3

With reference to Fig. 5.3, suggest why some parts of the molecule are double stranded.

.....

[2]

Fig. 5.4 shows a structure of the tryptophan (*trp*) operon of *Escherichia coli*, a prokaryote.

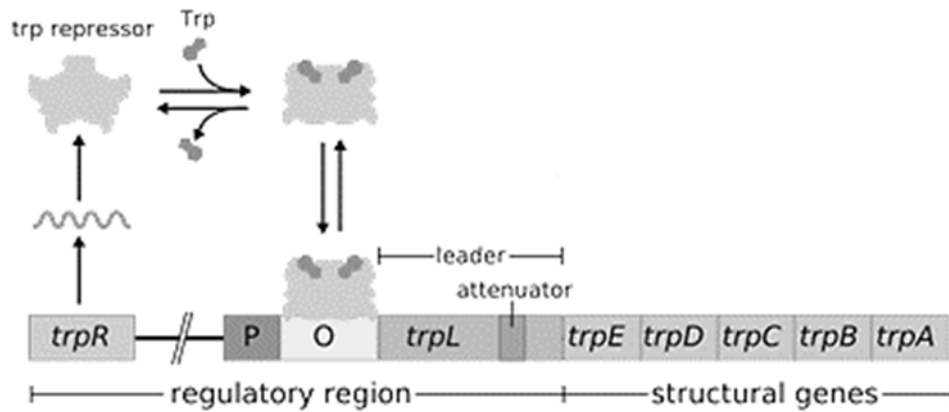


Fig. 5.4

(d) With reference to Fig. 5.4, explain the role of an operator in *trp* operon.

.....

[1]

(e) Explain the effect on the transcription of the operon if *trp* repressor protein is mutated.

.....

[2]

(f) Explain the significance of negative feedback in both inducible and repressible operons in prokaryotes.

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[2]

[Total: 10]

- 6** The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced in cells called melanocytes by the action of the enzyme tyrosinase. Low enzyme activity leads to the production of red pigment, while high enzyme activity brings about black pigment production.

The activity of the enzyme is increased when melanocyte stimulating hormone (MSH), a hormone that is normally present in healthy cattle, binds to a receptor coded for by the gene, **R**. No receptor is produced by the recessive allele, **r**.

The dominant allele of a second gene, **B**, codes for an inhibitor protein that binds to and blocks the MSH receptors coded for by **R**, thus preventing stimulation of tyrosinase activity in a melanocyte. The recessive allele, **b**, does not produce a functional protein.

- (a)(i)** State the name given to the interaction between the **R** and **B** gene loci.

.....[1]

- (ii)** Identify all possible genotype(s) for the black coat phenotype for the Norwegian cattle and explain how the alleles interact to give rise to the black colouration.

.....

.....

.....

.....[2]

- (b)** A red cow, with genotype **RRBB** is mated with a bull which is homozygous recessive at both gene loci. Draw a genetic diagram in the space below to show the expected genotypes and phenotypes and their ratios in the F_1 and F_2 generations.

7 Fig. 7.1 shows a flowchart showing the process of cellular respiration in a muscle cell.

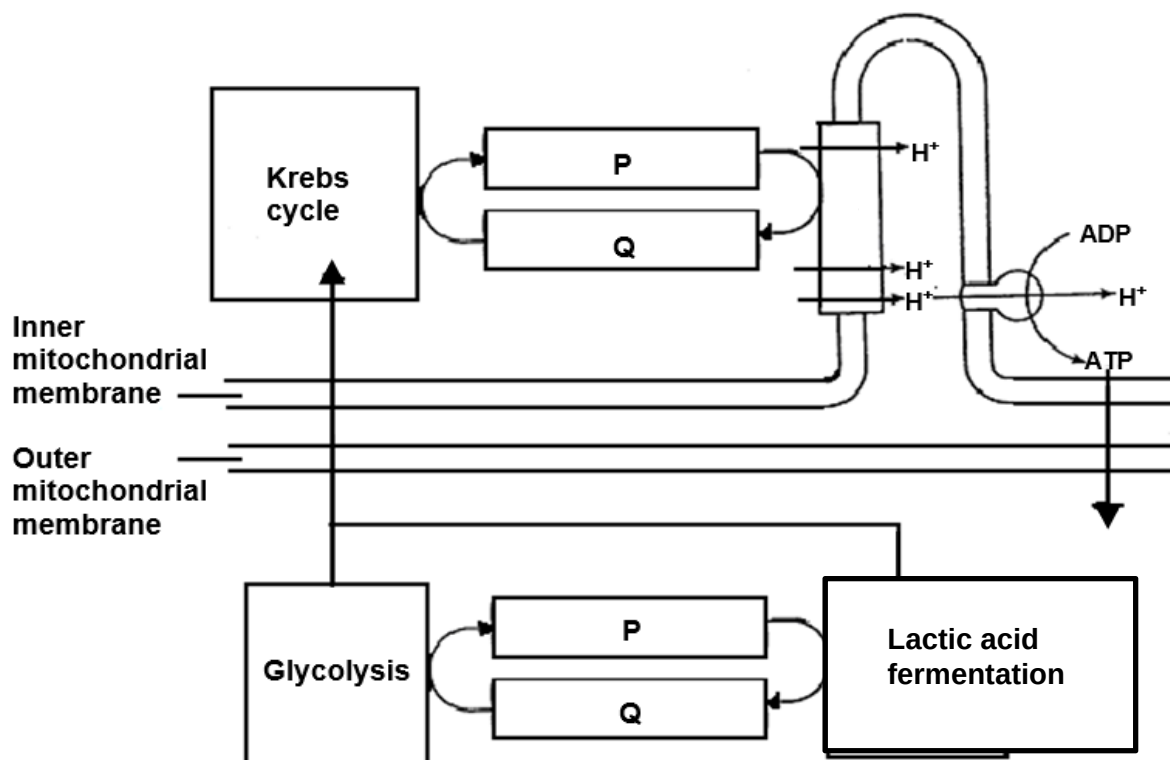


Fig. 7.1

- (a) P and Q plays an important role in ensuring that glycolysis always happen in either aerobic or anaerobic respiration process.

(i) Identify P and Q.

P:

Q: [1]

(ii) Explain the significance of P and Q in linking Krebs cycle to the formation of ATP in the cristae.

.....

 [2]

- (b) With reference to Fig. 7.1, explain why lactic acid fermentation occurs.

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 [2]

(c) Pyruvate from glycolysis has to be converted to acetyl CoA in a link reaction.

State the location where link reaction occurs.

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[1]

(d) Fig. 7.2 shows the Krebs cycle process.

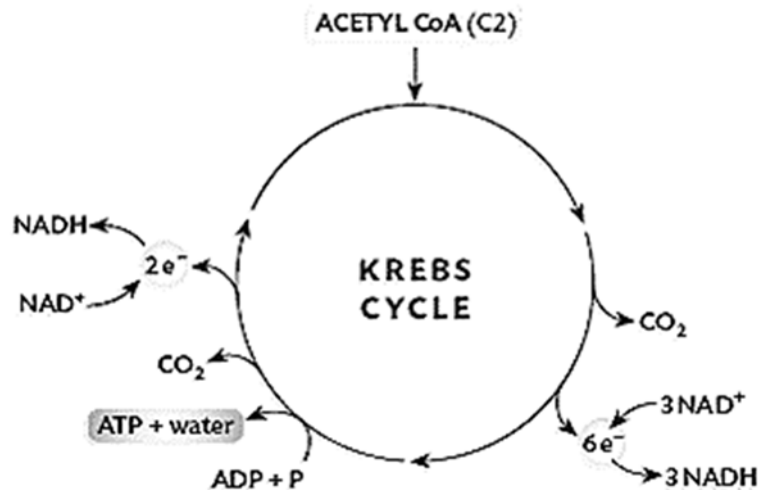


Fig. 7.2

Suggest a significance of acetyl-CoA in metabolism of *other* respiratory substrates.

.....

[2]

(e) In plant cells, on the other hand, has organelles where their cellular respiration synergizes with photosynthesis process. Calvin cycle, a light independent reaction, occurs in the stroma of chloroplast which plays an important role in photosynthesis.

Outline 2 ways in which Calvin cycle differs from Krebs cycle.

.....

[2]

[Total: 10]

- 8 Fig. 8.1 shows the distribution of the salamander *Ensatina* of the Pacific Coast region of the United States. Their collective range stretches approximately 1400km down from the Canadian coast, forking into a 'coastal arm' and an 'inland arm' at the dry, hot lowland region of the Central Valley; till the ranges finally meet again upon reaching the Southern Californian Coast. The first salamanders were hypothesized to have originated in the north.

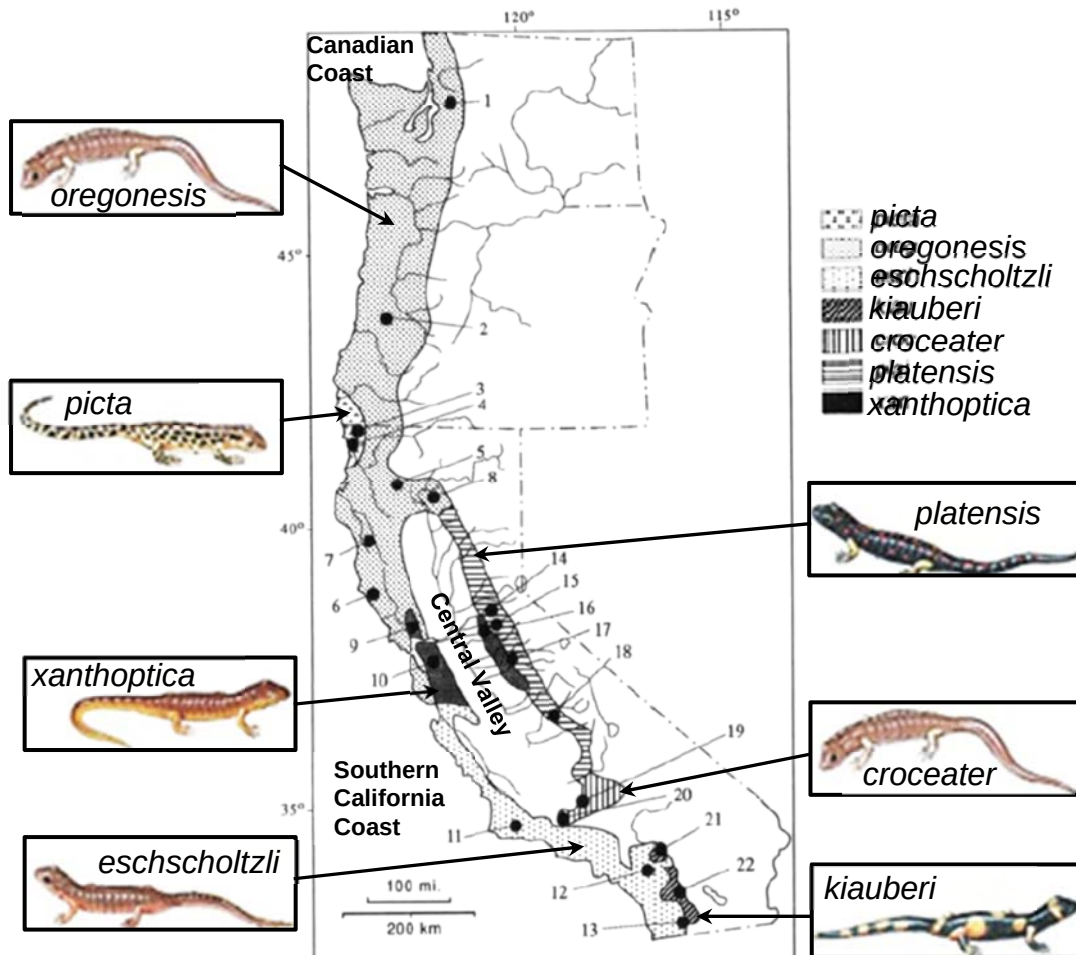


Fig. 8.1 Range of the salamander *Ensatina* of the Pacific Coast region of the United States. Numbered labels represent sample collection sites. <http://sysbio.oxfordjournals.org/> at North Carolina State University on March 25, 2014

- (a) Explain the role of natural selection in evolution.

.....[1]

E. picta, found exclusively on the eastern coast of California has a range of less than a hundred kilometers compared to its ancestor *E. oregonesis* which extends to an excess of a thousand kilometers.

- (b)** With reference to Fig. 8.1 suggest why the range of *E. picta* is so limited compared to *E. oregonesis*.

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.....[4]

It was later discovered that *E. oregonesis*, *E. picta* and *E. xanthoptica* were found to have fertile hybrids within the population; while further south *E. eschscholtzii*, *E. platensis*, *E. croceator* and *E. kiauberi* were incapable of interbreeding.

- (c)(i)** Discuss what conclusions can be derived from the mating study of the seven groups of Salamander.

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.....[2]

- (c)(ii)** Explain the application of a known species concept upon which your answer in **(c)(i)** is based upon.

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.....[2]

- (d)(ii)** What reliable technical method can be used to better support your answer in **(c)(i)**?

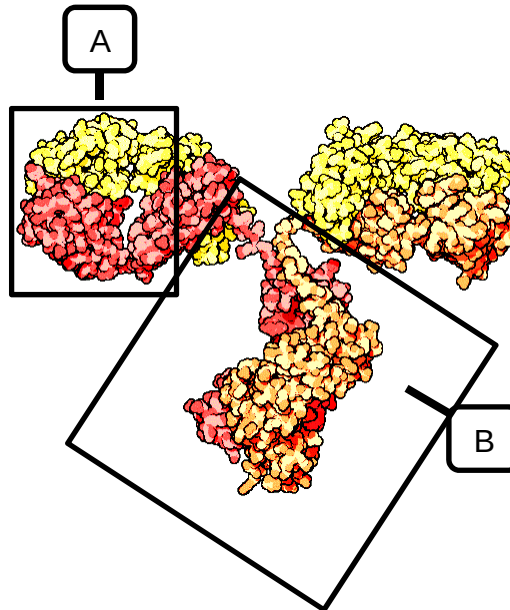
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.....[1]

[Total: 10]

- 9 Fig. 9.1 shows the globular structure of a B cell receptor IgM after the B cell has just left the lymph node.

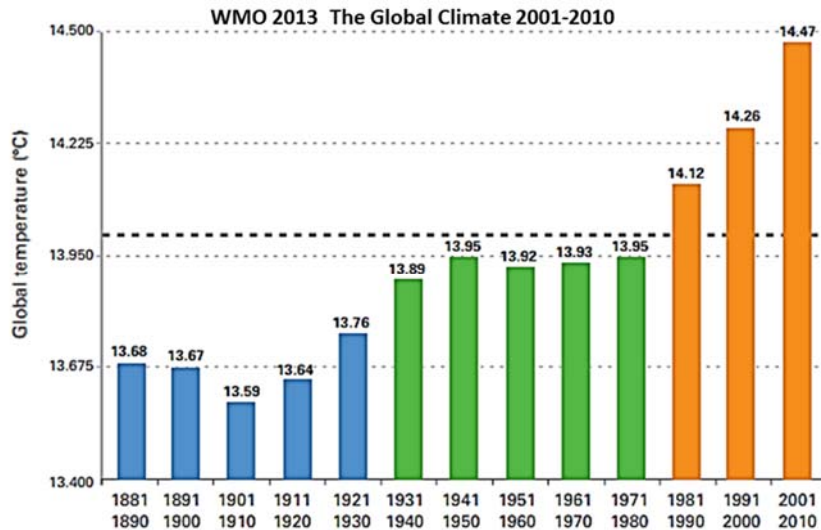
Fig. 9.1



- (a) State the developmental stage of these B cells as part of the cellular innate defense.
-[1]
- (b) With reference to Fig. 9.1, region A outlines a section of IgM that is involved in recognition. State and explain the process occurring in lymph node that this cell had undergone upon meeting a specific pathogen.
-[2]
- (c) With reference to Fig. 9.1, explain why region B is different from corresponding regions of other similar structures like IgD, IgG from the same type of B Cells.
-[2]
- (d) Explain how immunological memory is established and its significance.
-[3]
- (e) Type 1 diabetes is a chronic disease. In type 1 diabetes, beta cells (which make insulin) in the pancreas are destroyed by the body's immune system. Suggest at which stages of the innate immune system may be the cause for Type 1 diabetes, explain your answer.
-[2]

[Total: 10]

10 Fig. 10.1
combined
land and



shows the global
surface air
temperature over
sea.

Fig. 10.1

(a)(i) As seen in Fig. 10.1 global temperatures have been on a constant rise, give two gases and their respective relative global warming potentials, which would contribute to this global phenomenon.

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.....[2]

(a)(ii) With a named example, explain what is positive feedback in global warming.

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.....
.....
.....[2]

(b) Global warming also has had significant effects on the ranges of both animals and plants. Explain what some of these effects are.

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.....
.....[3]

Only 22% of Canadian bird species spend the whole year in Canada (Fig. 10.2A). Most others migrate to the United States (33%), to Central America, Mexico and the Caribbean (23%) or to South America (15%). Some travel to Europe or Asia or spend long periods of time at sea (7%).

A forty year study of these migratory species (Fig. 10.2B) has shown declines in species migrating to the United States, Mexico and very drastic declines in Brazil, South America.

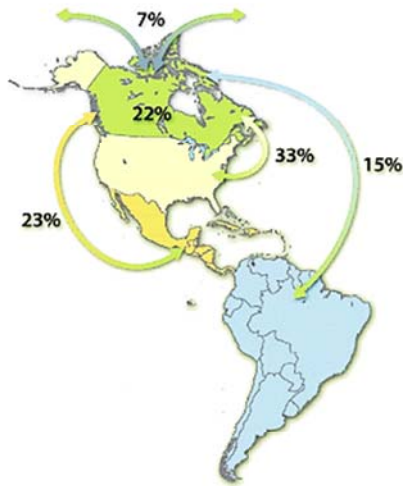


Fig.10.2A Destinations of Canada's migratory bird species

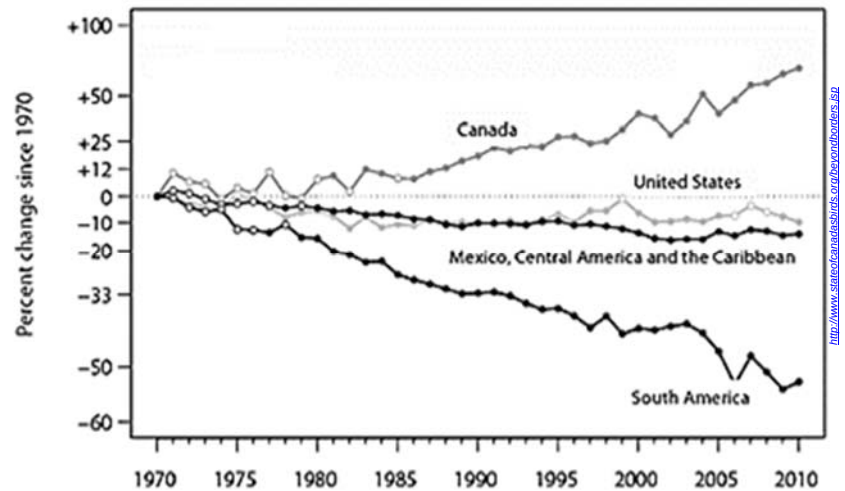


Fig.10.2B Indicators of average population status of species that primarily winter in four major

(c)(i) With reference to Fig. 10.2A&B, suggest what might be a possible direct anthropomorphic cause for the decline in Brazil, explain why.

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.....

.....[2]

(c)(ii) With reference to Fig. 10.2B, suggest a biological impact of the decline of migratory species in Brazil.

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.....[1]

[Total: 10]

END OF PAPER